Thomas Keith Roberts

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Profile Summary

Early-career Computational Neuroscientist and Data Scientist from the University of Birmingham. I'm a trained MEG operator with experience in high-level analysis, including source localization, speech-decoding and BCIs. I have created software for HPC clusters and behavioural studies. Dedicated to following high-quality scientific practices and communicating this information in a way that is digestible for non-technical stakeholders.

Experience

University Of Birmingham Sep 2025 — Present
Teaching Fellow *Birmingham, United Kingdom*

- Lead Seminars in Psychology and Research Methods to students with a variety of background including psychology, mathematics and biology.
- Taught data science and python to undergraduates and postgraduates this included statistical methods, computational methods, and data manipulation.
- Marking for undergraduate and postgraduate students including data analysis coding assignments and research proposals.
- Collaborated with senior lecturers to create materials to aid student in education of psychology and data science.

Centre for Human Brain Health May 2025 — Present
Researcher *Birmingham, United Kingdom*

- Analyzed trial-based face processing time-series neurological data capture via MEG using MNE Python and the FLUX pipeline.
- Built a CLI tool to help researchers on High Performance Computing Clusters (HPC) by automating stable defaults that are inline with CHBH best practices to improve accessibility and replicability of analysis.
- Presented high-quality literature to a small groups of researchers, developers and non-technical staff.
- Created documentation for Neureca Lab, including fixing out of date or incorrect information, managing research profiles and public facing information.

University of Birmingham Dec 2021 — Present
Ambassador *Birmingham, United Kingdom*

- Acted as a representive to the university to prosepective students, international visiters and members of the civil service.
- Presented to Schools about University Life and Neuroscience.
- Conducted tours around campus.

Education

University of Birmingham Birmingham, United Kingdom
Master of Science in Data Science and Cognitive Neuroimaging (Computational Neuroscience) Sep 2025 — Present

- Electrophysiology
- Computational and Mathematical Models
- Open Science Framework

Korea University Seoul, South Korea
Exchange Program Sep 2023 — Jul 2024

- Neural Engineering
- Systems Neuroscience
- Korean Language

University of Birmingham Birmingham, United Kingdom
Bachelor of Science in Psychology with a Year Abroad Sep 2021 — Jul 2025

- Psychology
- Neuroscience
- Data Science

Projects

Inner Speech Decoding using Magnetoencelography Sep 2025 — Present

MNE-Python | HPC | PyTorch (https://github.com/TommyTNeuro/non-invasive-meg-inner-speech-decoding)

- Analyzed source-localized MEG data measured from trial-based inner speech task using MNE-Python.
- Wrote a high-quality report and documentation on inner speech decoding.
- Used Open Science Framework to preregister project.

CHBH CLI May 2026 — Present

Python | Bash (https://github.com/chbh-opensource/chbh-cli)

- Conceptualized Idea to help beginners to Neuroimaging analysis on HPC clusters.
- Created packages for standard MEG analysis.
- Created documentation for tool.

Neuroimagers Handbook May 2025 — May 2026

Git | Python (https://github.com/chbh-opensource/chbh-on-bear)

- Helped with transfer of relevant material from CHBH wiki.
- Contributed to MEG section.
- Helped others contribute to documentation.

Invesigated Language Features on Spatial Attention Sep 2024 — May 2025

PsychoPy | Pandas | Matplotlib (https://github.com/TommyTNeuro/cancellation-task-stim-exp-pipeline)

- Created a Spatial Attention tasks including cancellation task using PsychoPy (Python) and Psychotoolbox (MATLAB).
- Conducted research using behavioral data including eye-tracking and computer-based tasks.
- Developed robust and replicable analysis pipeline that is applicable for a wide range of spatial spatial attention research.

Publications

For publications please see the publication page on my website

Skills

- **Research**: Trained MEG Operator, Behavioral Data Collection, Eye Tracking.
- **Programming Languages**: Python, Rust, MATLAB, C++, R, Bash
- **Analysis**: MNE-Python, PsychoPy, PyTorch, Pandas, Numpy, Matplotlib, Microsoft Suite
- **Tools**: Git, Linux, High Performance Computing, Documentation

Teaching and Writings

- Created the Hitchhiker's Guide to Data Science for Data Science for Brain and Behavior module (University of Birmingham)
- Created material for Social and Developmental Psychology
- Various Writing on Electrophysiology, Neuroscience and Psychology.

Memberships

- British Neuroscience Assocaiation
- FENS (Federation of European Neuroscience Societies)